

Physics 827: Homework Set No. 7

**Due date: Friday, November 18, 2011, 1:00pm
in PRB M2043 (Biao Huang's office)**

Total point value of set: 50 points

Problem 1 (5 pts.): Exercise 7.3.3 (Shankar, p. 196)

Problem 2 (10 pts.): Exercise 7.3.4 (Shankar, p. 196). No MATHEMATICA!

Problem 3 (10 pts.): Exercise 7.3.5 (Shankar, p. 196) No MATHEMATICA!

Problem 4 (15 pts.): Exercise 7.3.6 (Shankar, p. 197)

Additional questions: What is the ground state energy and wavefunction? What is the level spacing between energy eigenvalues? Show that your eigenfunctions are correctly orthonormalized!

Problem 5 (10 pts.): Exercise 7.3.7 (Shankar, p. 202)

Exercise 7.3.3 (p.196)

Given: $\psi(x) = \psi(-x)$ (even)

$\phi(x) = -\phi(-x)$ (odd)

• Then $\int_{-\infty}^{\infty} dx \phi(x) \psi(x) = \left(\int_{-\infty}^0 + \int_0^{\infty} \right) dx \phi(x) \psi(x)$

$$\stackrel{(x=-y)}{=} \int_{y=\infty}^0 (-dy) \underbrace{\phi(-y)}_{-\phi(y)} \psi(-y) + \int_0^{\infty} dx \phi(x) \psi(x)$$

$$= -\int_0^{\infty} dy \phi(y) \psi(y) + \int_0^{\infty} dx \phi(x) \psi(x) = 0$$

• Therefore, with $\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega x^2}{2\hbar}} = \text{even}$

$$\psi_1(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \underbrace{2\left(\frac{m\omega}{\hbar}\right)^{1/2} x}_{H_1\left(\sqrt{\frac{m\omega}{\hbar}}x\right)} e^{-\frac{m\omega x^2}{2\hbar}} = \text{odd}$$

being the wavefunctions for the ground and first excited state of the harmonic oscillator,

$$\langle \psi_0 | \psi_1 \rangle = \int_{-\infty}^{\infty} dx \underbrace{\psi_0^*(x)}_{\text{even}} \underbrace{\psi_1(x)}_{\text{odd}} = \int_{-\infty}^{\infty} dx \underbrace{\psi_0(x)}_{\text{even}} \underbrace{\psi_1(x)}_{\text{odd}} = 0 \quad \checkmark$$

• Further, with $\psi_2(x) = \left(\frac{m\omega}{\pi\hbar 16 \cdot (2!)^2}\right)^{1/4} \left(-2\left(1 - 2\frac{m\omega}{\hbar}x^2\right)\right) e^{-\frac{m\omega x^2}{2\hbar}}$

we find

$$\begin{aligned} \langle \psi_0 | \psi_2 \rangle &= -\sqrt{\frac{m\omega}{\pi\hbar}} \frac{1}{12} \int_{-\infty}^{\infty} dx \left(1 - 2\frac{m\omega}{\hbar}x^2\right) e^{-\frac{m\omega x^2}{\hbar}} \\ &= -\frac{1}{12\pi} \int_{-\infty}^{\infty} d\zeta (1 - 2\zeta^2) e^{-\zeta^2} = -\frac{1}{12\pi} \left(\sqrt{\pi} - 2 \cdot \frac{1}{2}\sqrt{\pi}\right) = 0 \quad \checkmark \end{aligned}$$

and $\langle \psi_1 | \psi_2 \rangle = -\sqrt{\frac{m\omega}{2\hbar}} \int_{-\infty}^{\infty} d\zeta \zeta (1 - 2\zeta^2) e^{-\zeta^2} = 0 \quad \checkmark$ by symmetric integration

Exercise 7.3.4 (p. 196)

I can write (7.3.26) more symmetrically:

$$\int_{-\infty}^{\infty} d\xi H_n(\xi) H_{n'}(\xi) e^{-\xi^2} = \delta_{nn'} \sqrt{\pi 2^{n+n'} n! n'!}$$

With this I compute

$$\langle n' | \hat{X} | n \rangle = \sqrt{\frac{m\omega}{\pi \hbar 2^{n+n'} n! n'!}} \int_{-\infty}^{\infty} dx x H_{n'}\left(\sqrt{\frac{m\omega}{\hbar}} x\right) H_n\left(\sqrt{\frac{m\omega}{\hbar}} x\right) e^{-\frac{m\omega x^2}{\hbar}}$$

$$= \sqrt{\frac{\hbar}{\pi m\omega} \frac{1}{2^{n+n'} n! n'!}} \int_{-\infty}^{\infty} d\xi \xi H_{n'}(\xi) H_n(\xi) e^{-\xi^2}$$

$$(7.3.25) = \sqrt{\frac{\hbar}{\pi m\omega} \frac{1}{2^{n+n'} n! n'!}} \int_{-\infty}^{\infty} d\xi \left(\frac{1}{2} H_{n'+1}(\xi) + n' H_{n'-1}(\xi) \right) H_n(\xi) e^{-\xi^2}$$

$$(7.3.26) = \sqrt{\frac{\hbar}{\pi m\omega}} \left\{ \frac{1}{2} \sqrt{\frac{2^{n+n'+1} n! (n'+1)!}{2^{n+n'} n! n'!}} \delta_{n'+1, n} + n' \sqrt{\frac{2^{n+n'-1} n! (n'-1)!}{2^{n+n'} n! n'!}} \delta_{n'-1, n} \right\}$$

$$= \sqrt{\frac{\hbar}{2m\omega}} \left\{ \sqrt{n'+1} \delta_{n'+1, n} + n' \sqrt{\frac{1}{n'}} \delta_{n'-1, n} \right\}$$

$$= \sqrt{\frac{\hbar}{2m\omega}} \left(\delta_{n', n+1} \sqrt{n+1} + \delta_{n', n-1} \sqrt{n} \right) \checkmark$$

Next,

$$\langle n' | \hat{P} | n \rangle = \sqrt{\frac{m\omega}{\pi \hbar 2^{n+n'} n! n'!}} \int_{-\infty}^{\infty} dx H_{n'}\left(\sqrt{\frac{m\omega}{\hbar}} x\right) e^{-\frac{m\omega x^2}{2\hbar}} \left(-i\hbar \frac{d}{dx}\right) H_n\left(\sqrt{\frac{m\omega}{\hbar}} x\right) e^{-\frac{m\omega x^2}{2\hbar}}$$

$$= -i\hbar \sqrt{\frac{m\omega}{\pi\hbar}} \frac{1}{2^{n+n'} n! n'!} \int_{-\infty}^{\infty} d\xi H_n(\xi) e^{-\xi^2/2} \frac{d}{d\xi} (H_n(\xi) e^{-\xi^2/2})$$

$$= -i \sqrt{\frac{m\omega\hbar}{\pi}} \frac{1}{n! n'! 2^{n+n'}} \int_{-\infty}^{\infty} d\xi H_n(\xi) e^{-\xi^2/2} \left[H_n'(\xi) e^{-\xi^2/2} - \xi H_n(\xi) e^{-\xi^2/2} \right]$$

$$(7.3.24) \quad = +i \sqrt{\frac{m\omega\hbar}{\pi}} \frac{1}{n! n'! 2^{n+n'}} \int_{-\infty}^{\infty} d\xi e^{-\xi^2} H_n(\xi) \left[2n H_{n-1}(\xi) - \xi H_n(\xi) \right]$$

$$(7.3.25) \quad = -i \sqrt{\frac{m\omega\hbar}{\pi}} \frac{1}{n! n'! 2^{n+n'}} \int_{-\infty}^{\infty} d\xi e^{-\xi^2} H_n(\xi) \left[2n H_{n-1}(\xi) - \frac{1}{2} H_{n+1}(\xi) - n H_{n+1}(\xi) \right]$$

$$= +i \sqrt{\frac{m\omega\hbar}{\pi}} \frac{1}{2^{n+n'} n! n'!} \int_{-\infty}^{\infty} d\xi e^{-\xi^2} \left[\frac{1}{2} H_n(\xi) H_{n+1}(\xi) - n H_n(\xi) H_{n-1}(\xi) \right]$$

$$= i \sqrt{m\omega\hbar} \left\{ \frac{1}{2} \sqrt{\frac{2^{n+1+n'} n'! (n+1)!}{2^{n+n'} n! n'!}} \delta_{n', n+1} - n \sqrt{\frac{2^{n-1+n'} n'! (n-1)!}{2^{n+n'} n! n'!}} \delta_{n', n-1} \right\}$$

$$= i \sqrt{\frac{m\omega\hbar}{2}} \left(\delta_{n', n+1} \sqrt{n+1} - \delta_{n', n-1} \sqrt{n} \right) \checkmark$$

Exercise 7.3.5 (p. 196)

$$\bullet \langle n | \hat{X} | n \rangle = |A_n|^2 \int_{-\infty}^{\infty} dx \underbrace{H_n^2\left(\sqrt{\frac{m\omega}{\hbar}} x\right)}_{\text{even}} \underbrace{e^{-\frac{m\omega x^2}{2\hbar}}}_{\text{odd}} \cdot x = 0 \checkmark$$

$$\bullet \langle n | \hat{P} | n \rangle = |A_n|^2 \int_{-\infty}^{\infty} dx H_n\left(\sqrt{\frac{m\omega}{\hbar}} x\right) e^{-\frac{m\omega x^2}{2\hbar}} \left(-i\hbar \frac{d}{dx}\right) H_n\left(\sqrt{\frac{m\omega}{\hbar}} x\right) e^{-\frac{m\omega x^2}{2\hbar}}$$

$$= |A_n|^2 (-i\hbar) \int_{-\infty}^{\infty} d\xi H_n(\xi) e^{-\xi^2/2} \left[H_n'(\xi) e^{-\xi^2/2} - \xi H_n(\xi) e^{-\xi^2/2} \right]$$

$$= (-i\hbar) |A_n|^2 \int_{-\infty}^{\infty} d\xi \left[\underbrace{\xi H_n^2(\xi)}_{\text{odd even}} - \underbrace{H_n' H_n}_{\text{even}} \right] e^{-\xi^2}$$

↙ (7.3.25)

$$= +i\hbar |A_n|^2 \int_{-\infty}^{\infty} d\xi 2n H_{n-1}(\xi) H_n(\xi) e^{-\xi^2}$$

(7.3.26)

$$= \sim \delta_{n,n-1} = 0 \checkmark$$

$$\bullet \text{ Hence } \langle \hat{X}^2 \rangle = (\Delta X)^2 + \langle \hat{X} \rangle^2 = (\Delta X)^2$$

$$\text{and } \langle \hat{P}^2 \rangle = (\Delta P)^2 + \langle \hat{P} \rangle^2 = (\Delta P)^2$$

$$\bullet \underbrace{\langle 1 | \hat{X}^2 | 1 \rangle} = \sqrt{\frac{m\omega}{\pi\hbar 4}} \int_{-\infty}^{\infty} dx x^2 4 \left(\frac{m\omega}{\hbar} x^2\right) e^{-\frac{m\omega x^2}{\hbar}}$$

$$= \frac{2}{\sqrt{\pi}} \frac{\hbar}{m\omega} \int_{-\infty}^{\infty} d\xi \xi^4 e^{-\xi^2} = \frac{2}{\sqrt{\pi}} \frac{\hbar}{m\omega} \left(\frac{d^2}{d\alpha^2} \int_{-\infty}^{\infty} d\xi e^{-\alpha \xi^2} \right)_{\alpha=1}$$

$$= \frac{2}{\sqrt{\pi}} \frac{\hbar}{m\omega} \underbrace{\left(-\frac{d}{d\alpha} \left(\frac{1}{2} \frac{\sqrt{\pi}}{\alpha^{3/2}} \right) \right)}_{\frac{3}{4} \sqrt{\pi}} = \frac{3}{2} \frac{\hbar}{m\omega} \checkmark$$

$$\begin{aligned}
 \cdot \langle 1 | \hat{P}^2 | 1 \rangle &= \sqrt{\frac{m\omega}{4\pi\hbar}} \int_{-\infty}^{\infty} dx \, 4\frac{m\omega}{\hbar} \left(x e^{-\frac{m\omega x^2}{2\hbar}} \right) \left(-\hbar^2 \frac{d^2}{dx^2} \right) \left(x e^{-\frac{m\omega x^2}{2\hbar}} \right) \\
 &= \frac{2}{\sqrt{\pi}} \left(\frac{m\omega}{\hbar} \right) \left(\frac{\hbar^2}{4} \right) \int_{-\infty}^{\infty} d\xi \, \xi e^{-\xi^2/2} \frac{d^2}{d\xi^2} \left(\xi e^{-\xi^2/2} \right) \\
 &= -\frac{2}{\sqrt{\pi}} (\hbar m\omega) \int_{-\infty}^{\infty} d\xi \, \xi e^{-\xi^2/2} \frac{d}{d\xi} \left(e^{-\xi^2/2} - \xi^2 e^{-\xi^2/2} \right) \\
 &= -\frac{2}{\sqrt{\pi}} (\hbar m\omega) \int_{-\infty}^{\infty} d\xi \, \xi e^{-\xi^2/2} \left(-\xi e^{-\xi^2/2} - 2\xi e^{-\xi^2/2} + \xi^3 e^{-\xi^2/2} \right) \\
 &= -\frac{2}{\sqrt{\pi}} (\hbar m\omega) \int_{-\infty}^{\infty} d\xi \, e^{-\xi^2} \left(-3\xi^2 + \xi^4 \right) = -\frac{2}{\sqrt{\pi}} (\hbar m\omega) \left(-3\frac{1}{2}\sqrt{\pi} + \frac{3}{4}\sqrt{\pi} \right) \\
 &= +\frac{3}{2} m\omega\hbar \quad \checkmark
 \end{aligned}$$

$$\Rightarrow (\Delta X \cdot \Delta P)_{(1)} = \frac{3}{2}\hbar > \frac{\hbar}{2}$$

uncertainty product.

Finally,

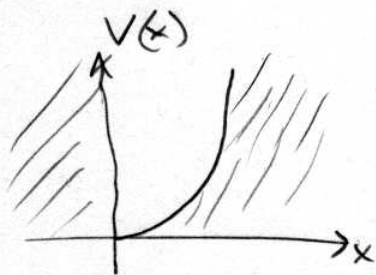
$$\cdot \langle 0 | \hat{X}^2 | 0 \rangle = \sqrt{\frac{m\omega}{\pi\hbar}} \int_{-\infty}^{\infty} dx \, x^2 e^{-\frac{m\omega x^2}{\hbar}} = \frac{1}{\sqrt{\pi}} \frac{\hbar}{m\omega} \int_{-\infty}^{\infty} d\xi \, \xi^2 e^{-\xi^2} = \frac{\hbar}{2m\omega}$$

$$\begin{aligned}
 \cdot \langle 0 | \hat{P}^2 | 0 \rangle &= \sqrt{\frac{m\omega}{\pi\hbar}} \int_{-\infty}^{\infty} dx \, e^{-\frac{m\omega x^2}{2\hbar}} \left(-\hbar^2 \frac{d^2}{dx^2} \right) e^{-\frac{m\omega x^2}{2\hbar}} \\
 &= -\frac{\hbar^2}{\sqrt{\pi}} \frac{m\omega}{\hbar} \int_{-\infty}^{\infty} d\xi \, e^{-\xi^2/2} \frac{d^2}{d\xi^2} \left(e^{-\xi^2/2} \right) \\
 &\quad \frac{d}{d\xi} \left(-\xi e^{-\xi^2/2} \right) \\
 &= -\frac{m\omega\hbar}{\sqrt{\pi}} \int_{-\infty}^{\infty} d\xi \, e^{-\xi^2/2} \left(-e^{-\xi^2/2} + \xi^2 e^{-\xi^2/2} \right) = \frac{m\omega\hbar}{\sqrt{\pi}} \int_{-\infty}^{\infty} d\xi \, (1-\xi^2) e^{-\xi^2} \\
 &\quad \sqrt{\pi} \left(1 - \frac{1}{2} \right) \\
 &= \frac{m\omega\hbar}{2}
 \end{aligned}$$

$$\Rightarrow \langle \Delta X \cdot \Delta P \rangle_{(0)} = \frac{\hbar}{2} \quad \checkmark \text{ Saturates the uncertainty relation}$$

Exercise 7.3.6 (p. 197)

$$V(x) = \begin{cases} \frac{m\omega^2}{2} x^2 & x > 0 \\ \infty & x \leq 0 \end{cases}$$



The infinite potential at $x \leq 0$ enforces the vanishing of all eigenfunctions at $x = 0$:

$$\psi_n(0) = 0$$

In the region $x > 0$, the eigenfunctions solve the same differential equation as for the harmonic oscillator:

$$\psi''(\xi) + (2\varepsilon - \xi^2)\psi(\xi) = 0 \quad \left(\xi = x\sqrt{\frac{m\omega}{\hbar}} \right)$$

This is solved by the ansatz

$$\psi(\xi) = u_n(\xi) e^{-\xi^2/2}$$

as before because the asymptotic behavior at $x \rightarrow 0^+$ and $x \rightarrow +\infty$ of the differential equation remains the same. The $u_n(\xi)$ satisfies the same power expansion

$$u_n(\xi) = \sum_{m=0}^{\infty} C_m \xi^m$$

with C_m satisfying the same recursion relation

$$C_{m+2} = C_m \frac{2m+1-2\varepsilon}{(m+2)(m+1)}$$

as before, since this relation follows from the structure of the differential equation, not from the boundary conditions.

Normalizability of the solution still enforces the truncation of the power series at finite $n \in \mathbb{N}_0$, because otherwise the solution $u_n(\xi)$ behaves like $e^{+\xi^2}$ at $\xi \rightarrow +\infty$. This yields the quantized energy eigenvalues

$$\varepsilon_n = \frac{2n+1}{2} \longleftrightarrow E_n = \left(n + \frac{1}{2}\right) \hbar \omega, \quad n = 0, 1, 2, \dots$$

as before, with corresponding solutions

$$\psi_n(\xi) = u_n(\xi) e^{-\xi^2/2} = \begin{cases} C_0 + C_2 \xi^2 + C_4 \xi^4 + \dots + C_n \xi^n & n \text{ even} \\ C_1 \xi + C_3 \xi^3 + \dots + C_n \xi^n & n \text{ odd} \end{cases} e^{-\xi^2/2}$$

However, the boundary condition $\psi_n(\xi=0)$ can now only be fulfilled if n is odd ($u_n(\xi) \xrightarrow{\xi \rightarrow 0} C_1 \xi + O(\xi^3) \rightarrow 0$) but not for even n where $u_n(\xi) \xrightarrow{\xi \rightarrow 0} C_0 \neq 0$.

(For $C_0 = 0$, all C_{n-2k} vanish, so C_0 must be $\neq 0$.)

So the boundary condition $\psi_n(0) = 0$ eliminates all solutions with even n , and the eigenvalue

Spectrum reduces to $(n=2k+1, k=0, 1, 2, \dots)$

$$E_k = (2k + \frac{3}{2}) \hbar \omega = \frac{4k+3}{2} \hbar \omega \quad k=0, 1, 2, \dots$$

with eigenfunctions

$$\psi_k(x) = \left(\frac{m\omega}{\pi \hbar 2^{4k} (2k+1)!} \right)^{1/4} H_{2k+1} \left(\sqrt{\frac{m\omega}{\hbar}} x \right) e^{-\frac{m\omega x^2}{2\hbar}} \theta(x) \quad *$$

$B_k = \sqrt{2} A_{2k+1} \quad k=0, 1, 2, \dots$

The ground state has energy

$$E_0 = \frac{3}{2} \hbar \omega$$

and the eigenfunction

$$\psi_0(x) = \left(\frac{m\omega}{\pi \hbar} \right)^{1/4} 2 \sqrt{\frac{m\omega}{\hbar}} x e^{-\frac{m\omega x^2}{2\hbar}} = 2 \left(\frac{m\omega}{\hbar} \right)^{3/4} \left(\frac{1}{\pi} \right)^{1/4} x e^{-\frac{m\omega x^2}{2\hbar}}$$

The energy splitting between eigenstates is

$$\Delta E = E_{k+1} - E_k = 2\hbar\omega$$

independent of k , so the energy spectrum has constant spacing of $2\hbar\omega$ (instead of $\hbar\omega$ for the harmonic oscillator).

*) These eigenfunctions are orthonormal on $(-\infty, \infty)$:

$$\int_{-\infty}^{\infty} \psi_k^*(x) \psi_k(x) dx = \int_0^{\infty} \psi_k^*(x) \psi_k(x) dx =$$

$$= B_k B_{k'} \int_0^{\infty} H_{2k+1} \left(\sqrt{\frac{m\omega}{\hbar}} x \right) H_{2k'+1} \left(\sqrt{\frac{m\omega}{\hbar}} x \right) e^{-\frac{m\omega x^2}{\hbar}} dx$$

$$= \frac{1}{2} B_k B_{k'} \int_0^{\infty} H_{2k+1} \left(\sqrt{\frac{m\omega}{\hbar}} x \right) H_{2k'+1} \left(\sqrt{\frac{m\omega}{\hbar}} x \right) e^{-\frac{m\omega x^2}{\hbar}} dx$$

$$(y=-x) + \frac{1}{2} B_k B_{k'} \int_0^{\infty} (-dy) \underbrace{H_{2k+1} \left(-\sqrt{\frac{m\omega}{\hbar}} y \right)}_{=-H_{2k+1} \left(\sqrt{\frac{m\omega}{\hbar}} y \right)} \underbrace{H_{2k'+1} \left(-\sqrt{\frac{m\omega}{\hbar}} y \right)}_{=-H_{2k'+1} \left(\sqrt{\frac{m\omega}{\hbar}} y \right)} e^{-\frac{m\omega y^2}{\hbar}} dy$$

(odd-order Hermite polynomials are odd fcts.)

$$= \underbrace{\frac{1}{2} B_k B_{k'}}_{A_{2k+1} A_{2k'+1}} \int_{-\infty}^{\infty} dx H_{2k+1} \left(\sqrt{\frac{m\omega}{\hbar}} x \right) H_{2k'+1} \left(\sqrt{\frac{m\omega}{\hbar}} x \right) e^{-\frac{m\omega x^2}{\hbar}}$$

$$= A_{2k+1} A_{2k'+1}$$

$$= \delta_{2k+1, 2k'+1} = \delta_{kk'}$$

$$\Rightarrow \boxed{\langle \psi_k | \psi_{k'} \rangle = \delta_{kk'}} \quad \checkmark$$

Exercise 7.3.7 (p. 202)

$$\left(\frac{\hat{P}^2}{2m} + \frac{m\omega^2}{2} \hat{X}^2 \right) |E\rangle = E |E\rangle$$

$$\int_{-\infty}^{\infty} \langle p | \frac{\hat{P}^2}{2m} + \frac{m\omega^2}{2} \hat{X}^2 | p' \rangle \langle p' | E \rangle dp' = E \langle p | E \rangle$$

$$\Rightarrow \frac{p^2}{2m} + \frac{m\omega^2}{2} (i\hbar)^2 \frac{d^2}{dp^2} \psi_E(p) = E \psi_E(p)$$

$$\Rightarrow \frac{d^2}{dp^2} \psi_E(p) + \frac{2}{m(\hbar\omega)^2} \left(E - \frac{p^2}{2m} \right) \psi_E(p) = 0$$

Introduce dimensionless momentum variable $\zeta \equiv p/p_0$:

$$\frac{d^2}{d\zeta^2} \psi(\zeta) + \frac{2E p_0^2}{m(\hbar\omega)^2} \psi - \frac{p_0^4}{(\hbar m \omega)^2} \zeta^2 \psi = 0$$

Choose $p_0 = \sqrt{\hbar m \omega}$: $\Rightarrow \boxed{\zeta = \frac{p}{\sqrt{\hbar m \omega}}}$

$$\frac{d^2}{d\zeta^2} \psi(\zeta) + \frac{2E \hbar m \omega}{m \hbar^2 \omega^2} \psi - \zeta^2 \psi = 0$$

Introduce $\varepsilon = \frac{E}{\hbar \omega}$ as before:

$$\boxed{\frac{d^2}{d\zeta^2} \psi(\zeta) + (2\varepsilon - \zeta^2) \psi(\zeta) = 0}$$

This is the same differential equation as before (7.3.8),

with solutions

$$\boxed{\psi_n(\zeta) = u_n(\zeta) e^{-\zeta^2/2} = B_n H_n(\zeta) e^{-\zeta^2/2}}$$

$\zeta = \frac{p}{\sqrt{\hbar m \omega}}$ differs from $\xi = x \sqrt{\frac{m\omega}{\hbar}}$ by the formal replacement $m\omega \rightarrow \frac{1}{m\omega}$ as we replace $x \rightarrow p$.

So all p -space solutions are obtained from the

x -space solutions by replacing $\xi \rightarrow \frac{p}{\sqrt{\hbar m \omega}}$ instead of

$\xi \rightarrow x \sqrt{\frac{m \omega}{\hbar}}$ and $A_n = \left(\frac{m \omega}{\pi \hbar 2^{2n} (n!)^2} \right)^{1/4}$ by $B_n = \left(\pi m \omega \hbar 2^{2n} (n!)^2 \right)^{-1/4}$.

$$\Rightarrow \psi_0(p) = \left(\frac{1}{\pi m \omega \hbar} \right)^{1/4} e^{-\frac{p^2}{2m(\hbar \omega)}} \quad \checkmark$$
